

Practice Quiz 11C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 11

Strong induction on a recursively defined sequence.

State the strong inductive hypothesis in full and cover both base cases.

1. Let $a_1 = 3$, $a_2 = 5$, and $a_n = 3a_{n-1} - 2a_{n-2}$ for $n \geq 3$. Prove that $a_n = 2^n + 1$ for every integer $n \geq 1$.

Solution We use strong induction. Let $n \geq 1$ and suppose

$$a_k = 2^k + 1 \quad \text{for every integer } k \text{ with } 1 \leq k < n.$$

If $n = 1$, then $a_1 = 3 = 2^1 + 1$. If $n = 2$, then $a_2 = 5 = 2^2 + 1$. If $n \geq 3$, the hypothesis applies at both $n - 1$ and $n - 2$, so

$$\begin{aligned} a_n &= 3a_{n-1} - 2a_{n-2} \\ &= 3(2^{n-1} + 1) - 2(2^{n-2} + 1) \\ &= 3 \cdot 2^{n-1} - 2^{n-1} + 1 = 2^n + 1. \end{aligned}$$

All cases are covered, so strong induction proves the formula for every $n \geq 1$. ■